Sprint 6 Report

Project Name: Personalized Study Planner

Team Name: Group 7

Team Members: Mustafa, Aashish, Minh, Mehga, Jari

Scrum Master: Mustafa

Sprint Duration: 2 Weeks

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1. Database Localization

Implemented Strategy

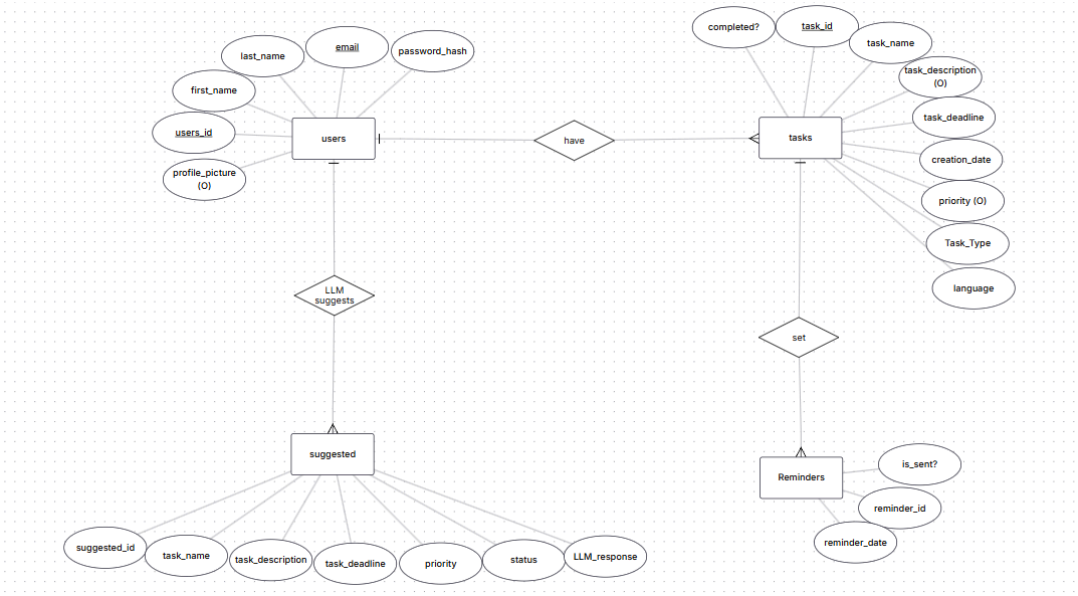
- Designed and implemented a localization strategy to extend support into the database layer.
- Updated the database schema and ERD to facilitate multilingual data handling.
- Configured the system to ensure proper UTF-8 character encoding and locale-specific data retrieval.

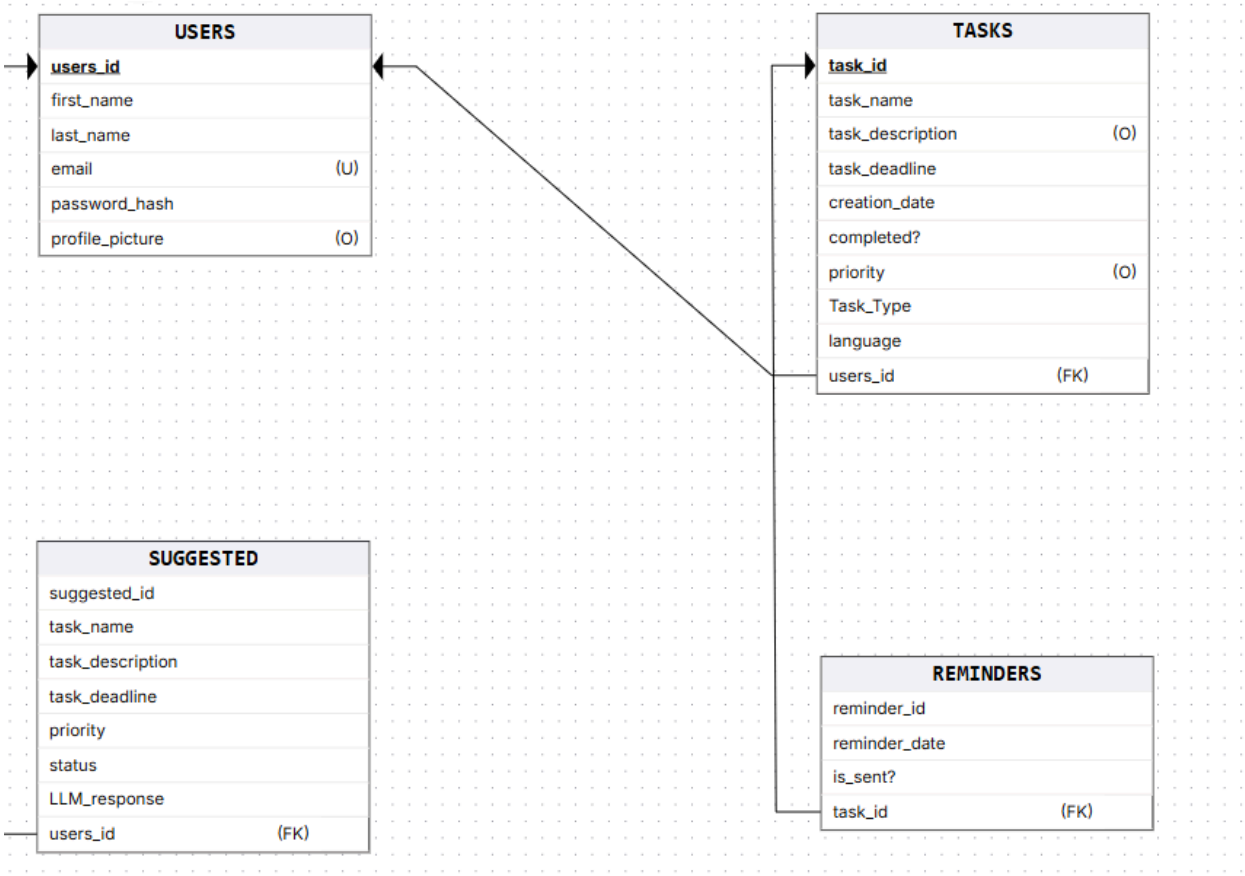
Deliverables

- Documentation of the chosen localization method added to the GitHub README.
- Row-based localization: Users may set the language of their choice from the given four options. **Language** will automatically be synched with the backend and stored in the database. When users create their tasks or generate tasks using LLM's they are stored in the DB and an additional column Language was added which shows the language of the user when creating the tasks.
- Additionally the LLM will respond in the language set by the user.

2. Architecture Design Documentation

- Updated system design artifacts, including the ER Diagram and UML models, to reflect localization changes.
- All design sketches and diagrams have been committed to the `/docs` folder in the GitHub repository.
- Screenshots of diagrams are attached below as well. You may notice the language attribute being added into the ER diagram and the relational schema complements the changes.


users
last_name
email
password_hash
first_name
users_id
profile_picture (O)
have
tasks
completed?
task_id
task_name
task_description (O)
task_deadline
creation_date
priority (O)
Task_Type
language
LLM suggests
suggested
suggested_id
task_name
task_description
task_deadline
priority
status
LLM_response
set
Reminders
is_sent?
reminder_id
reminder_date


USERS
users_id
first_name
last_name
email (U)
password_hash
profile_picture (O)
TASKS
task_id
task_name
task_description (O)
task_deadline
creation_date
completed?
priority (O)
Task_Type
language
users_id (FK)
SUGGESTED
suggested_id
task_name
task_description
task_deadline
priority
status
LLM_response
users_id (FK)
REMINDERS
reminder_id
reminder_date
is_sent?
task_id (FK)

```
study_planner=> select * from tasks where language='EN';
 task_id | completed |         created_at         | priority | status | task_deadline | task_description |         task_name         |         updated_at
     | user_id | google_event_id | from_llm_suggestion | suggested_task_id | shared_by_email | language
---------+-----------+----------------------------+----------+--------+---------------+------------------+---------------------------+----------------------
------+---------+-----------------+---------------------+-------------------+-----------------+----------
      36 | f         | 2026-04-11 15:54:19.955616 |        1 |      0 |               | change tables    | DB localization           | 2026-04-11 15:54:19.9
55616 |       8 |                 | f                   |                   |                 | EN
      38 | f         | 2026-04-11 16:36:44.751433 |        1 |      0 |               | implement it     | Implement DB localization | 2026-04-11 16:36:44.7
51433 |       9 |                 | f                   |                   |                 | EN
(2 rows)
```

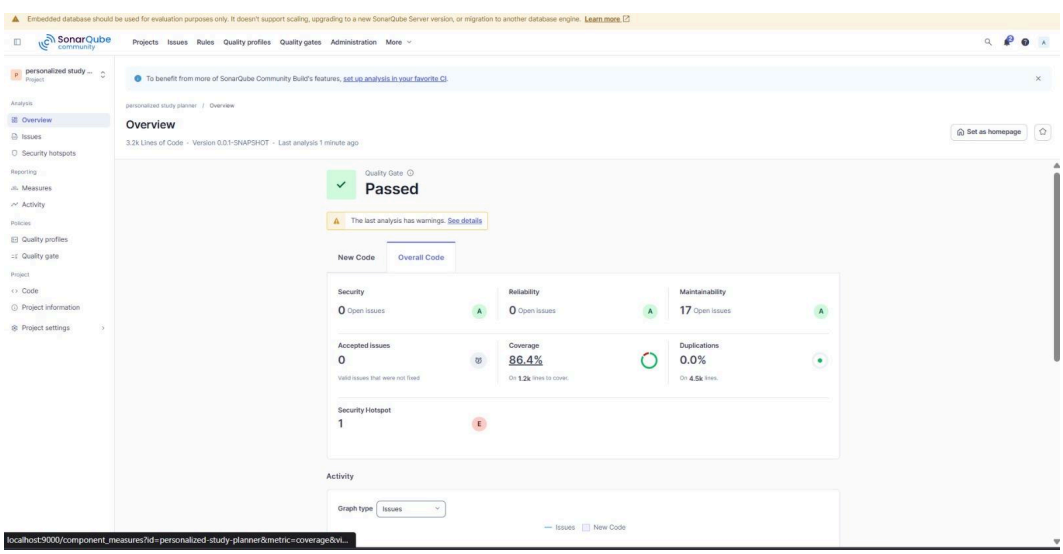
```
study_planner=> select * from tasks where language='FI';
 task_id | completed |         created_at         | priority | status | task_deadline |                    task_description                    |     task
_name      |         updated_at         | user_id | google_event_id | from_llm_suggestion | suggested_task_id | shared_by_email | language
---------+-----------+----------------------------+----------+--------+---------------+--------------------------------------------------------+---------
-----------+----------------------------+---------+-----------------+---------------------+-------------------+-----------------+----------
      37 | f         | 2026-04-11 15:56:33.607807 |        1 |      0 |               | Changing language and seeing if it works in the db or not | Changing
 language  | 2026-04-11 15:56:33.607807 |       8 |                 | f                   |                   |                 | FI
(1 row)
```

3. Statistical Code Review

Static Analysis Results

<u>Key Metrics Recorded:</u>

- **Cyclomatic Complexity:** Analyzed to identify overly complex and l
- **Lines of Code:** Monitored per method to ensure maintainability.
- **Redundancy:** Identified duplicate or unreachable code segments.
- **Code coverage:** 86.4 %
- **Duplications:** 0.0%
- **Lines of code:** 3204
- **Functions:** 216
- **LOC:** 14.77



4. Code Clean-Up and Refactoring

Refactoring Actions

- **Decomposition of Logic:** Complex functions were broken down into smaller, modular units to improve readability and maintainability.
- **Issue Resolution:** Addressed specific violations identified during the **Statistical Code Review** (SonarQube) to ensure the codebase adheres to quality standards.
- **Standardization:** Applied consistent naming conventions and formatting across the updated methods to meet Java coding conventions.
- **Documentation:** Added method-level comments and inline documentation to clarify the logic of the newly created smaller functions.
- **Stability Verification:** Re-run all unit tests to confirm that these structural changes did not introduce any regressions and that the system remains stable.
- **Validation**: Re-ran unit tests to confirm system stability and ensure no regressions occurred during refactoring.

```
[INFO] Results:
[INFO]
[WARNING] Tests run: 148, Failures: 0, Errors: 0, Skipped: 1
[INFO]
[INFO]
[INFO] --- jacoco:0.8.13:report (report) @ backend ---
```

5. Acceptance Test Planning

Test Design (Non-Execution) Established measurable success criteria based on original project requirements.

- **Test Case Categories:**
  - Functional Testing
  - Usability testing.
  - Performance and reliability testing.
- Mapped all test cases to corresponding user stories in a coverage matrix.
- Acceptance test reported added in github repo

6. Members Contributions

Team Member Name	Assigned Tasks	Time Spent (hrs)	In-class tasks
Mustafa	Scrum Master, Database Localization Plan, Design Documentation	20	Submitted
Aashish	Static Code Analysis & SonarQube Setup	20	Submitted
Minh	Code Refactoring & Naming Standardization	20	Submitted
Megha	Static Code Analysis & SonarQube Setup	20	Submitted
Jari	Acceptance Test Planning	20	Submitted